
TEMPO: Regime-Aware Operator Learning with Locally Adapted POD Bases

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Abstract

Operator learning methods typically approximate the solution map of a physical system with a single global model—an assumption that fails when solutions change character fundamentally across the parameter space.

We propose TEMPO: a framework that identifies latent dynamical regimes in snapshot data via Expectation-Maximization, builds a dedicated Neural-POD basis for each, and learns to route new inputs to the right regime at inference time. On multi-regime benchmarks, TEMPO achieves substantially lower errors than POD-DeepONet and vanilla DeepONet, with interpretable structure that reflects the underlying physics.

Keywords: Operator learning, DeepONet, Proper Orthogonal Decomposition, Expectation-Maximization, Gaussian mixture models, Regime discovery

1 Introduction

Operator learning addresses the approximation of mappings between infinite-dimensional function spaces, arising naturally in the solution of parametric partial differential equations (PDEs) governing physical systems [1]. This paradigm has demonstrated substantial promise across both theoretical analysis [1, 2] and practical applications including weather modeling, quantum systems, fluid and sound dynamics [3–6].

Among the most prominent architectures, Deep Operator Networks (DeepONet) [2] approximate operators via paired branch and trunk networks, where the branch encodes the input function and the trunk encodes the query location. The Fourier Neural Operator (FNO) [7] independently parameterizes the solution operator in the spectral domain, offering resolution-invariance and strong empirical performance. POD-DeepONet [8] improves upon standard DeepONet by replacing the learned trunk network with a fixed basis obtained from Proper Orthogonal Decomposition (POD) of training snapshots, yielding more compact and interpretable representations.

However, a single global POD basis is inherently limited for problems whose solutions exhibit qualitatively distinct dynamical regimes—such as solutions transitioning between laminar and turbulent behavior, or dispersive waves with parameter-dependent structure. In such settings, the optimal basis varies across the solution manifold, and a single linear subspace cannot represent the full solution diversity.

Several approaches enrich the POD basis to overcome this limitation. Sharma and Shankar [9] update the DeepONet trunk with multiple complementary bases—precomputed POD modes and a partition-of-unity mixture-of-experts (PoU-MoE) trunk, trained jointly with a single branch. On 2D and 3D problems with sharp spatial gradients, the POD-PoU ensemble achieves up to $2\times$ lower errors over POD-DeepONet [8] and $2\text{--}4\times$ over vanilla DeepONet.

Moreover, the linear structure of the POD basis is insufficient for strongly nonlinear solution manifolds. Mou et al. [10] propose Neural-POD, which constructs nonlinear orthogonal basis functions via iterative residual minimization analogous to Gram–Schmidt orthogonalization, representing each spatial mode and temporal component as neural networks trained sequentially on the residual variance.

Yet both approaches enrich the basis without identifying the underlying dynamical regimes present in the data. This idea has been pursued in the reduced order modeling (ROM) literature, where snapshot ensembles are partitioned into clusters and a separate local POD basis is constructed for each [11, 12]. More recently, Daniel et al. [13] extended this paradigm by clustering solution subspaces on the Grassmann manifold and training a deep neural network to select the appropriate local model from a dictionary at inference time. However, these methods operate offline on fixed snapshot sets, employ hard cluster assignments, and do not integrate regime discovery with operator learning in an end-to-end trainable architecture.

In this work, we propose **TEMPO** (*Regime-Aware Operator Learning with Locally Adapted POD Bases*). Our method integrates regime discovery directly into operator learning: a Gaussian mixture model over Neural-POD coefficient space identifies latent dynamical regimes via soft EM, constructing a dedicated locally adapted basis for each. At inference time, a learned gating network selects among regimes, producing predictions that are accurate across the full parameter space.

2 Problem Statement

The behavior of many physical systems is governed by a set of parameters $\kappa \in \mathcal{K} \subset \mathbb{R}^{d_\kappa}$: viscosity controls whether flow is laminar or turbulent, boundary forcing shapes wave patterns, material coefficients determine whether stress propagates smoothly or develops shocks. As κ varies across the parameter space $\mathcal{K}$, the corresponding solutions can change not just quantitatively but qualitatively, demanding fundamentally different representations in each regime. Predicting these solutions accurately and instantly, without re-running the underlying simulator, is the central problem we address.

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain and $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ a time interval.

We are given a dataset of N solution snapshots:

$$\mathcal{D} = \left\{ (u^{(i)}, \kappa^{(i)}, t^{(i)}, s^{(i)}) \right\}_{i=1}^N, \quad s^{(i)}(\cdot) = s(\cdot, t^{(i)}; u^{(i)}, \kappa^{(i)}) \in L^2(\Omega), \quad (1)$$

where $u^{(i)} \in \mathcal{U} = L^2(\Omega)$ is the input function, $\kappa^{(i)} \in \mathcal{K}$ the physical parameter, $t^{(i)} \in \mathcal{T}$ the observation time, and $s^{(i)}$ the corresponding solution discretized on N_y quadrature points $\{(x_j, w_j)\}_{j=1}^{N_y}$ with norm $\|f\|^2 = \sum_j w_j f(x_j)^2$.

The dataset $\mathcal{D}$ captures $\mathcal{G}$ through a finite collection of snapshots; the underlying operator is defined as follows.

Definition 1 (Solution operator). *The solution operator*

$$\mathcal{G} : \mathcal{U} \times \mathcal{K} \rightarrow L^2(\Omega \times \mathcal{T})$$

maps each pair (u, κ) to the full space-time solution: $\mathcal{G}(u, \kappa) = s(\cdot, \cdot; u, \kappa)$.

The operator $\mathcal{G}$ is unknown and expensive to evaluate directly. The *operator learning problem* is to construct $\hat{\mathcal{G}} \approx \mathcal{G}$ from $\mathcal{D}$ alone, generalizing to unseen $(u, \kappa) \notin \mathcal{D}$ without solving the PDE.

The multi-regime assumption

When solutions change character fundamentally across $\mathcal{K}$, the optimal low-dimensional representation differs from regime to regime, and no single global basis captures the full solution diversity. We formalize this observation via a latent variable model over the snapshot ensemble.

Definition 2 (Generative model). *Each snapshot arises from one of M latent regimes. The regime label $z_i \in \{1, \dots, M\}$ is unobserved and drawn from a categorical prior,*

$$z_i \sim \text{Categorical}(\pi_1, \dots, \pi_M), \quad \pi_m \geq 0, \quad \sum_m \pi_m = 1. \quad (2)$$

Conditioned on $z_i = m$, the snapshot is distributed as

$$(s^{(i)} \mid z_i=m) \sim \mathcal{N}(\hat{s}_m^{(i)}, \sigma^2 W^{-1}), \quad (3)$$

where $W = \text{diag}(w_1, \dots, w_{N_y})$ is the quadrature weight matrix, $\sigma^2 > 0$ controls the softness of regime assignments, and $\hat{s}_m^{(i)}$ is the regime- m approximation to $s^{(i)}$.

The generative model leaves $\hat{s}_m^{(i)}$ unspecified. We require only that it admits a *separable low-rank form*:

$$\hat{s}_m^{(i)}(x) = \phi_0^{(m)}(x; \kappa^{(i)}) + \sum_{p=1}^{P_m} \phi_p^{(m)}(x; \kappa^{(i)}) \psi_p^{(m)}(t^{(i)}; \kappa^{(i)}), \quad (4)$$

where $\phi_p^{(m)}(\cdot; \kappa)$ are spatial basis functions and $\psi_p^{(m)}(\cdot; \kappa)$ temporal coefficients, both conditioned on κ . This form encompasses POD [8], kernel PCA [14], and learned neural bases [10]; in this work we adopt the latter.

Definition 3 (Regime approximator). *The parameters of regime m are $\Phi_m = \{\phi_p^{(m)}, \psi_p^{(m)}\}_{p=0}^{P_m}$, where $\phi_0^{(m)}(\cdot; \kappa)$ is a parameter-dependent mean, $\{\phi_p^{(m)}\}_{p \geq 1}$ spatial basis networks, $\{\psi_p^{(m)}\}_{p \geq 1}$ temporal coefficient networks, and $P_m \leq P_{\max}$ the adaptive mode count.*

The learning problem decomposes into two phases: discovering the latent regime structure from $\mathcal{D}$ *offline*, then training a deployable operator for fast inference *online*.

Phase 1: Offline regime discovery

Since the regime labels z_i in (3) are unobserved, we recover the bases $\{\Phi_m\}$, mixture weights π , and noise level σ^2 by maximizing the observed-data log-likelihood:

$$(\Phi^*, \pi^*, \sigma^{2*}) = \arg \max_{\Phi, \pi, \sigma^2} \sum_{i=1}^N \log \sum_{m=1}^M \frac{\pi_m \det(W)^{1/2}}{(2\pi\sigma^2)^{N_y/2}} \exp\left(-\frac{\|s^{(i)} - \hat{s}_m^{(i)}\|^2}{2\sigma^2}\right). \quad (5)$$

The solution yields soft regime assignments $\gamma_{im}^* = p(z_i=m \mid s^{(i)}, \Phi^*, \pi^*, \sigma^{2*})$ that encode the discovered regime structure of $\mathcal{D}$ and serve as supervision for Phase 2.

Phase 2: Online operator learning

With bases $\{\Phi_m^*\}$ and responsibilities $\{\gamma_{im}^*\}$ frozen, we train a *GatingNet* and M *BranchNets* that map a new input (u, κ) to a weighted combination of local regime predictions. The training objective balances three goals: reconstruction accuracy on $\mathcal{D}$, consistency of the learned gating with the offline regime structure $\{\gamma_{im}^*\}$, and diversity across regimes to prevent collapse. The precise architecture and loss are given in Section 7.

The quality of the final approximation is measured by the mean relative L^2 error on held-out test snapshots:

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\|s^{(i)} - \hat{s}(\cdot; u^{(i)}, \kappa^{(i)})\|}{\|s^{(i)}\|}. \quad (6)$$

The algorithm solves Phase 1 via EM and Phase 2 via gradient descent; details are in Section 7.

3 Theory

4 Computational Experiments

We evaluate TEMPO on parametric PDE benchmarks, verifying that (i) the offline phase discovers interpretable regime structure, (ii) the learned bases faithfully represent regime-specific solution features, and (iii) TEMPO achieves lower reconstruction error than single-basis baselines.

Datasets. We use the PDEBench benchmark [15] across three benchmark problems.

1D Burgers’ equation.

$$u_t + u u_x = \nu u_{xx}, \quad x \in (-1, 1), \quad t \in (0, 2], \quad (7)$$

with periodic boundary conditions and random initial conditions sampled from a fixed distribution. The viscosity $\nu \in \{0.001, 0.01, 0.1, 1.0\}$ spans qualitatively distinct regimes: smooth diffusive profiles at $\nu = 1.0$ and near-discontinuous shock structures at $\nu = 0.001$. Each value provides $N = 9,500$ trajectories on a uniform grid of $N_x = 1024$ spatial points and $N_t = 101$ time steps, yielding 38,000 trajectories in total.

2D Darcy Flow.

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in (0, 1)^2, \quad (8)$$

with zero Dirichlet boundary conditions

Methods. We compare five methods. As baselines we use DeepONet [2], POD-DeepONet [8], and Neural-POD [10]. TEMPO is evaluated in two variants depending on the basis used in the offline phase: **TEMPO(POD)** constructs regime-specific bases via weighted SVD, and **TEMPO(NeuralPOD)** replaces them with Fourier NeuralPOD bases [10], which capture nonlinear and parameter-dependent solution structure.

Metrics. We report the mean relative L^2 error

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{\|s^{(i)} - \hat{s}(\cdot; u^{(i)}, \kappa^{(i)})\|}{\|s^{(i)}\|}, \quad (9)$$

together with RMSE and MSE, computed per viscosity value and aggregated over all ν .

Selection of M . The number of regimes is chosen from $M \in \{2, 3, 4\}$ by jointly minimising BIC on the EM-GMM fit and mean assignment entropy $H = -\frac{1}{N} \sum_{i,m} \gamma_{im} \log \gamma_{im}$; lower entropy indicates more confident regime assignments. Table 1 reports both criteria; $M = 3$ achieves the best trade-off and is used throughout.

Table 1: Regime selection on 1D Burgers’

M	BIC	Entropy
2	—	—
3	—	—
4	—	—

Regime structure. Figures 1 and 2 visualise the latent structure discovered in the offline phase. The t-SNE and UMAP projections show that EM-GMM regime assignments do not align with individual viscosity values, confirming that the discovered regimes reflect solution geometry rather than the physical parameter directly. The GMM fit in POD coefficient space (α_1, α_2) demonstrates that regime separation is well-captured in the low-dimensional projection used for clustering.

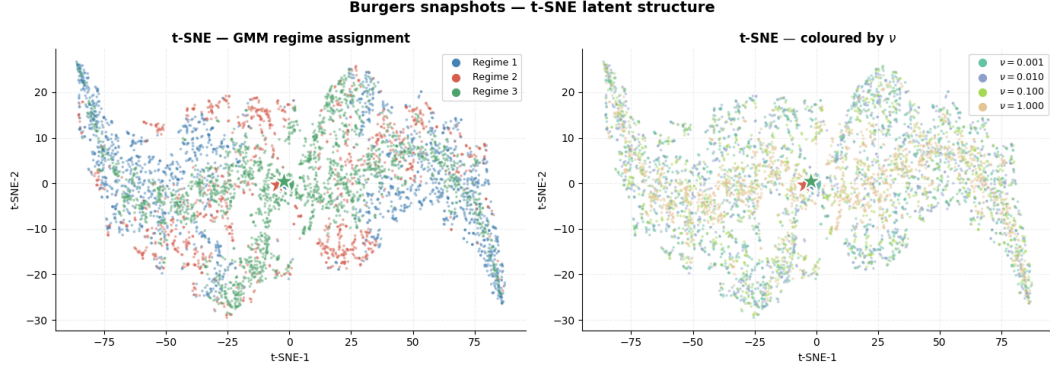


Figure 1: t-SNE projection of Burgers' snapshots ($N = 38,000$, $\nu \in \{0.001, 0.01, 0.1, 1.0\}$). EM-GMM regime assignments ($M = 3$);

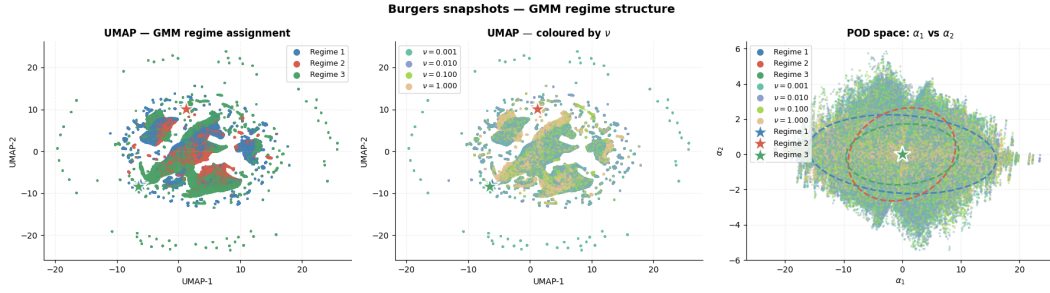


Figure 2: TEMPO regime structure ($M = 3$). *Left*: UMAP coloured by regime assignment. *Centre*: same projection coloured by ν ; each branch spans multiple viscosity values. *Right*: GMM fit in POD coefficient space (α_1, α_2) ; stars indicate centroids μ_m .

Basis quality. Figure 3 shows the training dynamics of the Fourier NeuralPOD basis for a single regime. The residual MSE per mode and the MSE reduction per mode confirm that modes 1–2 capture the dominant share of the variance, with diminishing contributions beyond mode 4. Figure 4 shows the resulting basis functions alongside representative snapshot reconstructions: for large ν the modes are smooth and global, while for small ν they are sharply localised near the shock front, demonstrating the parameter-adaptivity of the NeuralPOD basis.

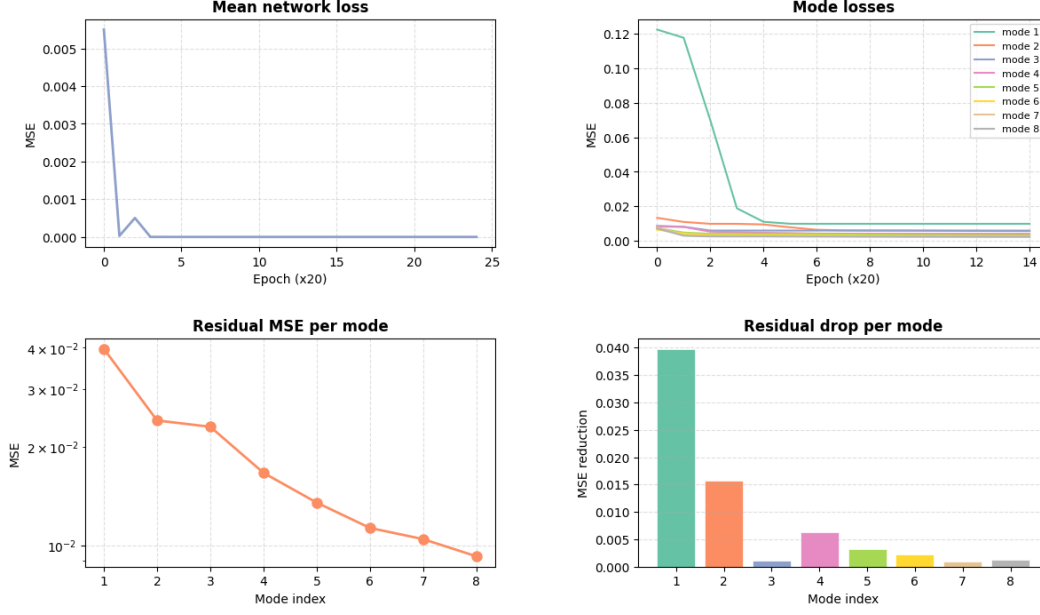


Figure 3: Fourier NeuralPOD training dynamics on 1D Burgers’

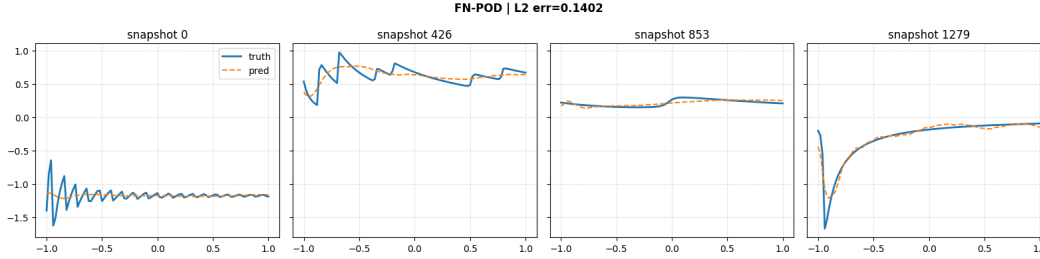


Figure 4: Learned Fourier NeuralPOD basis functions and representative reconstructions for 1D Burgers’.

Reconstruction accuracy. Table 2 reports accuracy for all methods, averaged over all viscosity values. Figure 5 shows space–time heatmaps of representative predictions and point-wise absolute errors.

Table 2: Reconstruction accuracy on the 1D Burgers’ test set.

Method	Rel. L^2	RMSE	MSE
DeepONet [2]	—	—	—
POD-DeepONet [8]	—	—	—
Neural-POD [10]	—	—	—
TEMPO(POD) (<i>ours</i>)	—	—	—
TEMPO(NeuralPOD) (<i>ours</i>)	—	—	—

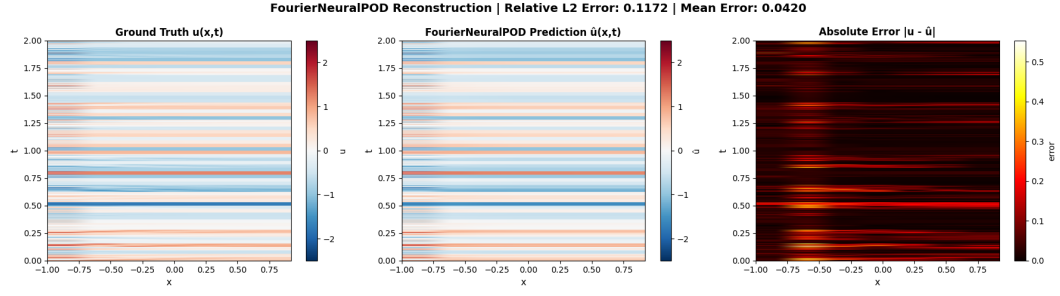


Figure 5: Space-time reconstruction at $\nu = 1.0$: ground truth (left), TEMPO(NeuralPOD) prediction (centre), and point-wise absolute error (right).

TODOOO

METHOD OR STATEMENT????????????????

Definition 4 (TEMPO predictor). *Given input function u sampled at sensor locations and parameter κ , the TEMPO prediction at query point $x \in \Omega$ is*

$$\hat{s}(x; u, \kappa) = \sum_{m=1}^M w_m(u, \kappa) \left[\Phi_0^{(m)}(x; \kappa) + \sum_{k=1}^{P_m} b_{m,k}(u, \kappa) \Phi_k^{(m)}(x; \kappa) \right], \quad (10)$$

where $w_m(u, \kappa) \in [0, 1]$, $\sum_m w_m = 1$, are the GatingNet outputs, and $b_{m,k}(u, \kappa)$ are the Branch-Net expansion coefficients for regime m

Online

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \left\| s^{(i)} - \hat{s}(\cdot; u^{(i)}, \kappa^{(i)}) \right\|^2, \quad (11)$$

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{i,m} w_m^{(i)} \log \frac{w_m^{(i)}}{\gamma_{im}^*}, \quad (12)$$

$$\mathcal{L}_{\text{ent}} = -\frac{1}{N} \sum_{i,m} w_m^{(i)} \log w_m^{(i)}, \quad (13)$$

combined as

$$(W^*, B^*) = \arg \min_{W, B} \mathcal{L}_{\text{online}}, \quad \mathcal{L}_{\text{online}} = \mathcal{L}_{\text{data}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}} - \lambda_H \mathcal{L}_{\text{ent}}, \quad (14)$$

Outline. The rest of the paper is organized as follows...

5 Related Work

Topic #1. TODO

Topic #2. TODO

6 Preliminaries

6.1 General notation

In this section, we introduce the general notation used in the rest of the paper and the basic assumptions.

6.2 Assumptions

TODO

7 Method

8 Experiments

To verify the theoretical estimates obtained, we conducted a detailed empirical study...

9 Discussion

TODO

10 Conclusion

TODO

References

- [1] Nikola B. Kovachki, Samuel Lanthaler, and Andrew M. Stuart. Operator learning: Algorithms and analysis. *arXiv*, 2024.
- [2] Lu Lu, Pengzhan Jin, and George Em Karniadakis. Deeponet: Learning nonlinear operators for identifying differential equations based on the universal approximation theorem of operators. *arXiv*, 2019.
- [3] Jaideep Pathak, Shashank Subramanian, Peter Harrington, Sanjeev Raja, Ashesh Chattopadhyay, Morteza Mardani, Thorsten Kurth, David Hall, Zongyi Li, Kamyar Azizzadenesheli, Pedram Hassanzadeh, Karthik Kashinath, and Anima Anandkumar. Fourcastnet: A global data-driven high-resolution weather model using adaptive fourier neural operators. *arXiv*, 2022.
- [4] Marco Olivieri, Xenofon Karakostas, Mirco Pezzoli, Fabio Antonacci, Augusto Sarti, and Efrén Fernandez-Grande. Physics-informed neural network for volumetric sound field reconstruction of speech signals. *EURASIP Journal on Audio, Speech, and Music Processing*, 2024: 17, 2024. doi: 10.1186/s13636-024-00366-2.
- [5] Himanshu Dave, Leo Cotteleer, and Alessandro Parente. Physics-informed non-intrusive reduced-order modeling of parameterized dynamical systems. *Computer Methods in Applied Mechanics and Engineering*, 443, 2025. doi: 10.1016/j.cma.2025.118045.
- [6] Kris Tucker, Amit Kiran Rege, Conor Smith, Claire Monteleoni, and Tameem Albash. Hamiltonian learning using machine learning models trained with continuous measurements. *arXiv*, 2025.
- [7] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *arXiv*, 2020.
- [8] Lu Lu, Xuhui Meng, Shengze Cai, Zhiping Mao, Somdatta Goswami, Zhongqiang Zhang, and George Em Karniadakis. A comprehensive and fair comparison of two neural operators (with practical extensions) based on fair data. *arXiv*, 2022.
- [9] Ramansh Sharma and Varun Shankar. Ensemble and mixture-of-experts deeponets for operator learning. *arXiv*, 2025.
- [10] Changhong Mou, Binghang Lu, and Guang Lin. Neural-pod: A plug-and-play neural operator framework for infinite-dimensional functional nonlinear proper orthogonal decomposition. *arXiv*, 2026.
- [11] David Amsallem, Matthew J. Zahr, and Charbel Farhat. Nonlinear model order reduction based on local reduced-order bases. *International Journal for Numerical Methods in Engineering*, 92(10):891–916, 2012. doi: 10.1002/nme.4371.
- [12] Martin Hess, Alessandro Alla, Annalisa Quaini, Gianluigi Rozza, and Max Gunzburger. A localized reduced-order modeling approach for PDEs with bifurcating solutions. *Computer Methods in Applied Mechanics and Engineering*, 351:379–403, 2019. doi: 10.1016/j.cma.2019.03.050.
- [13] Thomas Daniel, Fabien Casenave, Nissrine Akkari, and David Ryckelynck. Model order reduction assisted by deep neural networks (ROM-net). *Advanced Modeling and Simulation in Engineering Sciences*, 7(1):16, 2020. doi: 10.1186/s40323-020-00153-6.
- [14] Hamidreza Eivazi, Stefan Wittek, and Andreas Rausch. Nonlinear model reduction for operator learning. *arXiv*, 2024.
- [15] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Dan MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. Pdebench: An extensive benchmark for scientific machine learning. *arXiv*, 2022.

A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO